

NORTHWEST ARKANSAS, AR, USA

WMO: 723436

Lat: 36.283N

Lon: 94.300W

Elev: 1287

StdP: 14.03

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53922

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.1	15.9	0.5	5.9	14.2	5.5	7.6	21.0	25.5	51.5	23.5	51.0	7.6	350	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	96.0	74.1	92.7	74.7	90.3	74.4	78.1	88.6	77.1	87.5	75.9	86.2	8.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	139.0	83.8	73.9	132.4	82.5	72.8	127.4	81.3	42.5	88.6	41.3	87.3	40.2	86.3	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	20.0	18.1	DB	2.9	98.9	6.3	3.7	-1.6	101.5	-5.2	103.6	-8.7	105.7	-13.3	108.3
			WB	2.0	80.3	6.0	2.1	-2.4	81.8	-5.9	83.1	-9.3	84.2	-13.6	85.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	36.5	39.5	49.4	57.9	66.0	74.9	78.5	77.7	70.6	58.9	48.4	39.1	
	DBStd	17.09	11.18	11.35	10.26	8.63	7.48	5.00	4.90	5.27	6.58	8.69	9.37	9.64	
	HDD50	1448	435	327	137	30	3	0	0	0	0	22	138	356	
	HDD65	3962	882	716	487	236	81	2	0	1	24	230	501	802	
	CDD50	4447	18	32	119	267	499	747	883	859	618	297	89	19	
	CDD65	1490	0	1	5	24	112	300	419	395	192	39	3	0	
	CDH74	14534	0	5	37	200	895	2832	4429	4013	1771	333	19	0	
	CDH80	6036	0	0	2	25	200	1094	2095	1900	654	65	1	0	
Wind	WSAvg	7.8	8.8	9.1	9.6	9.7	8.0	7.0	5.8	5.7	5.9	7.3	8.5	8.3	
Precipitation	PrecAvg	46.4	2.8	2.4	3.7	4.5	5.7	5.0	3.7	3.0	4.8	3.7	4.0	3.0	
	PrecMax	65.0	7.4	6.1	10.3	8.2	10.2	10.7	11.3	6.4	11.0	9.4	9.0	6.5	
	PrecMin	34.4	0.3	0.3	0.7	1.9	1.9	1.0	1.0	0.2	0.4	1.4	0.5	0.5	
	PrecStd	7.5	1.7	1.6	2.2	2.0	2.0	2.4	2.1	1.5	2.4	1.9	2.6	1.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.5	72.4	78.2	83.2	88.1	95.3	99.7	100.9	92.9	85.1	76.7	68.2	
		MCWB	57.0	60.3	62.8	65.5	70.4	74.1	73.1	74.0	72.7	70.4	63.2	59.5	
	2%	DB	64.1	67.1	73.6	79.1	84.6	91.6	96.1	96.6	90.2	81.7	71.1	63.9	
		MCWB	55.9	56.3	60.6	63.9	69.9	74.9	73.9	74.4	72.8	68.3	60.5	56.6	
	5%	DB	60.1	63.0	70.0	75.8	82.1	88.9	92.8	92.8	87.3	77.8	67.8	59.4	
		MCWB	51.2	54.4	58.2	62.6	69.7	74.3	75.2	74.3	72.1	65.0	58.9	52.3	
	10%	DB	54.5	58.5	66.2	72.7	79.4	86.3	90.1	89.9	83.8	73.5	64.1	55.2	
		MCWB	47.2	50.7	56.0	60.8	68.0	73.2	75.3	74.1	70.5	63.7	56.3	49.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.5	62.9	65.4	69.4	74.7	78.8	80.1	78.9	77.2	72.9	67.3	62.8
			MCDB	65.4	69.2	74.0	78.4	82.6	89.2	90.5	89.3	88.4	81.8	73.8	66.3
2%		WB	58.5	59.5	62.9	67.1	72.8	77.1	78.5	77.6	75.0	70.9	63.5	58.5	
		MCDB	63.0	64.6	71.0	75.1	81.1	87.4	88.8	88.7	85.5	78.9	68.0	62.5	
5%		WB	52.7	56.1	60.5	65.2	71.3	75.6	77.3	76.4	73.5	68.7	60.6	54.4	
		MCDB	57.7	61.0	67.4	72.5	79.5	85.3	87.4	87.5	83.3	74.1	65.9	58.0	
10%		WB	47.7	51.2	58.3	63.3	69.7	74.4	76.2	75.2	72.2	66.2	57.7	50.5	
		MCDB	53.1	57.0	64.8	70.4	77.4	83.7	86.2	86.3	81.0	71.4	62.8	54.3	
Mean Daily Temperature Range	5% DB	MDBR	20.1	20.3	21.7	21.9	20.2	19.8	20.6	20.9	21.9	21.6	21.4	19.2	
		MCDBR	26.5	25.7	26.0	24.8	22.3	22.0	23.8	24.9	24.3	24.8	25.0	24.3	
		MCWBR	18.1	17.5	15.3	13.8	11.0	8.8	7.6	7.6	9.5	12.7	16.0	17.4	
	5% WB	MCDBR	21.9	21.8	21.0	20.6	19.4	19.4	19.9	20.8	20.8	19.5	21.5	21.0	
		MCWBR	17.9	17.2	14.3	13.5	10.7	9.0	8.5	8.5	9.5	11.4	16.8	17.6	
Clear-Sky Solar Irradiance	taub	0.302	0.305	0.339	0.377	0.420	0.421	0.442	0.443	0.386	0.343	0.312	0.291		
	taud	2.485	2.487	2.423	2.328	2.259	2.280	2.275	2.249	2.376	2.481	2.521	2.536		
	Ebn at Noon	285	298	295	287	275	273	267	264	275	279	279	283		
	Edh at Noon	26	29	34	40	43	43	42	43	36	29	25	23		
All-Sky Solar Radiation	RadAvg	780	948	1276	1609	1785	2040	2024	1837	1567	1164	838	664		
	RadStd	69	134	76	92	152	181	124	125	118	150	93	84		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page